

3D-Trajectory and Phase-Shift Design for RIS-Assisted UAV Systems Using Deep Reinforcement Learning

Haibo Mei , Kun Yang , Senior Member, IEEE, Qiang Liu , Member, IEEE, and Kezhi Wang , Senior Member, IEEE

Abstract—Unmanned aerial vehicle (UAV) can effectively work as temporary base station or access point in the air to transfer/receive data to/from ground terminals (GTs). However, UAV-GT links might be blocked by ground obstacles, like buildings in urban area, leading to a poor performance on data transferring rate. To address this problem, reconfigurable intelligent surface (RIS), as a promising technique, can intelligently reflect the received signals between UAV and GT to significantly enhance the communication quality. Under this deployment of RIS-assisted UAV, we intend to jointly optimize the 3D-space of the UAV and the phase-shift of the RIS to maximize the data transferring rate of the UAV, while minimizing the UAV propulsion energy. The joint problem is non-convex in its original form and difficult to be timely solved by using traditional method, like successive convex approximation (SCA). Therefore, to facilitate the online decision making to this joint problem, we leverage deep reinforcement learning (DRL) to learn the near-optimal solution, and the well known Double Deep Q-Network (DDQN) and Deep Deterministic Policy Gradient (DDPG) algorithms are utilized. Numerical results show that DRL can effectively improve the energy-efficiency performance of the RIS-Assisted UAV system, compared with benchmark solutions.

Index Terms—Reconfigurable intelligent surface (RIS), intelligent reflecting surface (IRS), deep reinforcement learning (DRL), 3D-trajectory, unmanned aerial vehicle (UAV).

I. INTRODUCTION

CURRENTLY, unmanned aerial vehicle (UAV) has been widely used in various wireless communication systems [1]. UAV can work flexibly in 3D-space to fascinate the air-ground wireless communication by simultaneously adjusting its horizontal and vertical positions in a timely manner, which is impossible for conventional territorial wireless networks [2]. Due to the high mobility, such UAV-assisted wireless network is particularly suitable for the on demand deployment in emergency scenarios, like catastrophic fields [3], where UAV is mostly used as the temporary base station or access point in the air to transfer/receive data to/from ground terminals (GTs).

UAV-assisted wireless network has attracted enormous research interests, and numbers of works have been proposed. The work in [1] discussed three typical use cases of UAV-assisted wireless communications including: UAV-aided ubiquitous coverage, UAV-aided relaying and UAV-aided information dissemination and data collection. In [4], the authors gave a tutorial overview of the recent advances in UAV communications, with an emphasis on how to integrate UAVs into the forthcoming fifth-generation (5G) and future cellular networks. In [5], it considered uplink communications in a multi-UAV enabled multi-user system under a new wireless network architecture of coordinate multipoint (CoMP). In [6], the authors worked on the UAV-enabled radio access network and aimed to minimize the UAV periodic flight duration or mission completion time, satisfying the target data requirement of each ground unit via a joint UAV trajectory and communication resource allocation design approach. In [7]–[9], it discussed UAV working as cellular users to extend the coverage of the network, which involves UAV trajectory optimization, interference management, resource management, etc. Moreover, the works in [10]–[12] studied that UAV faces major limitations caused by its constrained size, weight and power (SWAP), while working in the wireless networks. The SWAP constraints limit the working time of the UAV that consumes most of its on-board power on propulsion during flight and depletes its battery quickly. To address these problems, existing works tried to improve the UAV performance in the network via 3D-trajectory design [10], resource allocation [11], cache deployment [12], etc. Those works leveraged convex optimization [13] and deep reinforcement learning (DRL) [14]

Manuscript received July 27, 2021; accepted December 26, 2021. Date of publication January 19, 2022; date of current version March 15, 2022. This work was supported in part by the Natural Science Foundation of China under Grant 62132004, in part by the Chinese MOST Major Research and Development Project under Grant 2021YFB2900204, in part by the Sichuan Major R&D Projects under Grants 2021YFG0363 and 22QYCX0168, in part by the UESTC Yangtze Delta Region Research Institute-Quzhou under Grant 2021D003, and in part by the EU H2020 Project COSAFE under Grant GA-824019. The review of this article was coordinated by Prof. Jun Won Choi. (Corresponding author: Qiang Liu.)

Haibo Mei and Qiang Liu are with the School of Communication and Information Engineering, University of Electronic Science and Technology of China, Chengdu 611731, China (e-mail: haibo.mei@uestc.edu.cn; liuqiang@uestc.edu.cn).

Kun Yang is with the Yangtze Delta Region Research Institute (Quzhou) and School of Information and Communication Engineering, University of Electronic Science and Technology of China, Chengdu 324000, China, and also with the School of Computer Science and Electronic Engineering, University of Essex, Colchester CO4 3SQ, U.K. (e-mail: kunyang@essex.ac.uk).

Kezhi Wang is with the Department of Computer and Information Sciences, Northumbria University, Newcastle upon Tyne NE1 8ST, U.K. (e-mail: kezhi.wang@northumbria.ac.uk).

Digital Object Identifier 10.1109/TVT.2022.3143839

to solve correlated joint optimization problems to maximally extend the UAV abilities to assist the wireless communication.

Despite the advantages, in practical applications, the UAV-GT links are highly possible to be blocked by the territorial obstacles, like the buildings in urban area, causing signal attenuation and low data transferring rate between the UAV and GT [15]. This problem will escalate when UAVs flying in low-altitude in complicated environment. According to our best knowledge, only a few works considered such blocking problem of the low-altitude UAV. In [16], the authors provided analytical approach to optimizing the altitude of the Low-altitude aerial platforms (LAPs) platforms to provide maximum radio coverage on the ground. Following the analytical model in [16], the authors of [17] evaluated the air-ground communication throughput considering low-altitude UAV, and proposed a solution to minimize the total UAV energy consumption, including both propulsion energy and communication related energy, while satisfying the communication throughput requirement of each ground unit. Even though existing works can improve the blocked air-ground communication via trajectory optimization etc, the performance improvement however is marginal, given the UAV SWAP constraints.

To this problem, reconfigurable intelligent surface (RIS) has been recently proposed to improve the communication quality of UAV wireless networks. A RIS is a meta-surface equipped with integrated electronic circuits that can be programmed to alter an incoming electromagnetic field in a customizable way. Each RIS unit is implemented by reflect-arrays that employ varactor diodes or other Micro-Electrical-Mechanical Systems, and whose resonant frequency is electronically controlled [18]. In [19], the authors proposed realistic power consumption model to tackle the energy efficiency problem of RIS-assisted wireless communication. The authors of [20] have considered using RIS, as large antenna-array systems for data-transmissions with multiple single-antenna autonomous terminals, like the IoT end devices. The works in [21] considered RIS-aided single-cell wireless system, and solved the correlated problems to minimize the total transmit power at the access point. The authors of [22] evaluated the performance of an large-scale antenna system assisted by RIS, by formulating a tight upper bound of the ergodic spectral and investigate the effect of the phase shifts on the ergodic spectral efficiency in different propagation scenarios. Considering mobile edge computing, the authors of [23] studied the beneficial role of RISs in MEC-enabled wireless communication systems. To deal with the similar problem as [21], the authors of [24] investigated the joint design of transmit beamforming matrix at the base station and the phase shift matrix at the RIS, by leveraging recent advances in DRL. Overall, RIS has already been proved to be a promising technology to improve the performance of conventional territorial cellular systems, and the energy and spectral efficiencies of the cellular systems can be effectively reached by the help of RIS. In theory, RIS could help UAV wireless networks to overcome the signal blocking problem of the air-ground wireless communication.

The RIS-assisted UAV however is still in its infancy stage, and not comprehensively studied yet. The authors of [25]–[28] and [29] have done the pioneer works. In [25], it studied a UAV equipped with one RIS serves as a passive relay to

assist the communication between the users and the base station while considered the minimum secure data rate received by the eavesdropper. However, it is difficult to practically deploy the RIS that is carried on-board by the UAV and hardly ensures constant signal redirection. In [26], a UAV-assisted RIS symbolic radio system was designed, where the UAV acts to help the RIS transmit its own symbolic information to the base station. The symbolic information is carried by the on/off states of the RIS. In contrast to [25] and [26], RIS is most likely to be deployed on the surface of territorial buildings to redirect the signals within UAV and GTs or base stations. This will help to improve the air-ground communication of the UAV networks, and save the energy consumption of the UAV on flight propulsion to reach an energy-efficient performance. Under this deployment, the authors of [27] jointly designed UAV's trajectory, RIS scheduling, and communication resource allocation for the RIS-assisted UAV OFDMA system. In [28], it considered secure UAV communications assisted by RIS, and a joint design on UAV's trajectory, RIS's passive beamforming, and transmit power of the transmitters was introduced. Similarly, the joint optimization on UAV's trajectory and RIS's passive phase-shift was studied in [29]. The authors of [28] and [29] pointed out that the 3D-trajectory of UAV and the phase-shift of RIS are highly correlated in the RIS-assisted UAV system. UAV will adopt the 3D-trajectory not only considering the locations of GTs but also RIS with adjustable phase shift. Additionally, the phase shift of the RIS can be adapted with respect to the 3D-trajectory of the UAV and the timely locations of the GTs to reach the alignment of different signals for maximum data transferring rate. Nevertheless, those works in [28] and [29] tried to find the close-form solution to the joint trajectory-phase-shift problem of RIS-assisted UAV, using method like successive convex approximation (SCA). The proposed solutions thus involve high computation cost and hardly to be real-time deployed in an online manner, given the UAV system state always dynamically changing. Against this problem, the works in [30] and [31] were proposed to utilize DRL algorithms including Double-Deep Q-learning (DDQN) and Deep-Deterministic Policy Gradient (DDPG) to solve the joint UAV trajectory and passive beamforming problem in real-time. However, both of the two works only considered optimizing horizontal UAV trajectory thus failed to fully explore the mobility of the UAV. Those works in [30] and [31] didn't consider the passive phase-shift of the RIS with uniform planer array (UPA) neither but a uniform linear array (ULA) instead, which does not follow the real situations. Also the work in [30] converted the continuous phase-shift of the RIS in to discrete states in the DDQN algorithm, causing the beamforming control of the RIS in low accuracy.

In this paper, to maximize the energy-efficiency of the RIS-assisted UAV, we jointly optimize the 3D-trajectory of the UAV and the phase-shift of the RIS in an online manner using DRL. We both implement the DDQN and DDPG algorithms to fully explore the mobility of the UAV in 3D-space and reach the passive phase-shift of the RIS in a high accuracy. These two DRL algorithms then can support the RIS-assisted UAV providing satisfied data rate to each UEs and having minimized propulsion energy consumption. The performances of the DDQN

and DDPG algorithms are also compared. Generally, the primary contributions of this paper are:

- 1) We study the optimization problem of the RIS-assisted UAV to try to reach an energy-efficiency performance. We comprehensively consider the UAV propulsion energy consumption in 3D-space and the passive phase-shift of the RIS with UPA. This work thus intends to maximize the system gains of the UAV-assisted wireless network, by exploring the full ability of the UAV and RIS.
- 2) We utilize DRL to overcome the computation and complex issues of traditional method, like SCA, while solving the joint trajectory-phase-shift problem of the RIS-assisted UAV. Two realistic DDQN and DDPG algorithms are provided to convert the UAV trajectory optimization either into a discrete or continuous learning problem, which can easily converge to obtain the sub-optimal solution to the joint problem.
- 3) We realize the passive phase-shift of the RIS along the DRL process. The RIS passive phase-shift is implemented with respect to the online learnt 3D-trajectory of the UAV and the timely locations of the GTs to reach the alignment of different signals for maximum data transferring rate. This passive phase-shift therefore will enable the RIS to realize more effective signal redirection to improve the air-ground wireless communications.
- 4) A comprehensive validation work will be provided to justify the proposed solution of this paper, and the numerical results proof that the DRL based solution outperforms, as compared to those benchmark solutions. We also compare and analyze the performances the DDQN and DDPG algorithms with respect to the simulation data.

The remainder of this paper is organized as follows. In Section II, we describe the system model and formulate the problem. In Section III, we present the solution to the optimization problem. In Section IV, simulation results and analysis are presented. In Section V, we give conclusions and future work. In addition, the main notations used in this paper are summarized in Table I.

Other Notations: In this paper, $\mathbb{C}^{M \times N}$ denotes the set of $M \times N$ complex vectors. $\mathbb{R}^{1 \times N}$ denotes the array with N elements. $\text{diag}\{\cdot\}$ denotes the diagonalization operation. $(\cdot)^T$ denotes the transpose operation. $\mathbb{E}[\cdot]$ denotes the expectation operation. $\|\cdot\|$ denotes the determinant operation.

II. SYSTEM MODEL AND PROBLEM FORMULATION

A. System Model

In this paper, we consider a typical RIS-assisted UAV system as shown in Fig. 1, where a single UAV is deployed to serve GTs. Assume that the area of interest (AoI) is discretized into L cells with equal size. Also assume that $L_i^c = [x_i, y_i]^T \in \mathbb{R}^{2 \times 1}$ is the coordinate of the centre of cell i ; x_s and y_s are the distances of two adjacent cells in x-axis and y-axis respectively. Then the horizontal location of the UAV at time slot n can be denoted by $L_n^u \in \mathcal{L}$, where $\mathcal{L} \triangleq \{1, 2, \dots, L\}$, $n \in \mathcal{N} \triangleq \{1, \dots, N\}$ and N is the total number of time slots. Note that L_o^u and L_f^u are the centres of the pre-decided initial and final locations of the UAV. Then the horizontal trajectory of the UAV can

TABLE I
MAIN NOTATIONS

Notation	Definition
N, K, L	Number of Time slots, GTs, and cells of the AoI
x_s, y_s	Distances of adjacent cells in x-axis and y-axis
h_s	Distance of each height level of the UAV
$L_i^c = [x_i, y_i]^T$	The coordinate of the centre of cell i in AoI
L_n^u	The horizontal location of the UAV in time slot n
H_n^u	The vertical location of the UAV in time slot n
H	Level count of the UAV altitude
$h_{\min}, h_{\max}$	Minimal and Maximal height of the UAV
t_n^u	Duration of time slot n
$t_{\min}, t_{\max}$	Minimal and Maximal duration of a time slot
τ	UAV overall flight time
$V_{\max}^h, V_{\max}^v$	UAV maximal horizontal and vertical velocities
v_n^h, v_n^v	UAV horizontal and vertical velocities
e_n^{uav}	UAV propulsion energy in time slot n
P_0, P_1	The blade profile and induced power of the UAV
P_2	The descending/ascending power of UAV
U_{tip}	Tip speed of the rotor blade of the UAV
v_0	The mean rotor induced velocity in hover
d_0, s	Fuselage drag ratio and rotor solidity of the UAV
ρ, G	Air density and rotor disc area of the UAV
$M_c \times M_r$	Number of PRUs of RIS
θ_{m_r, m_c}	Phase shift inserted at PRU (m_r, m_c)
d_c, d_r	Distance of adjacent PRUs in column and row
w_R, z_R	The location of the first PRU of RIS
$w_k = [x_k, y_k]^T$	The location of GT k
B, P	Bandwidth, transmitting power of the UAV
$p_{k,n}$	The LoS probability between UAV and GT k
a, b	The constants depending on the wireless environment
σ	The noise variance
ξ	The path loss at the reference distance 1m
g_n^{ur}	The channel gain of UAV-RIS link
g_k^{rg}	The channel gain between RIS and GT k
$g_{k,n}^{\text{avg}}$	The average channel gain at GT k
$r_{k,n}$	The received data rate at GT k in time slot n
$c_{k,n}$	0-1 variable on GT scheduling

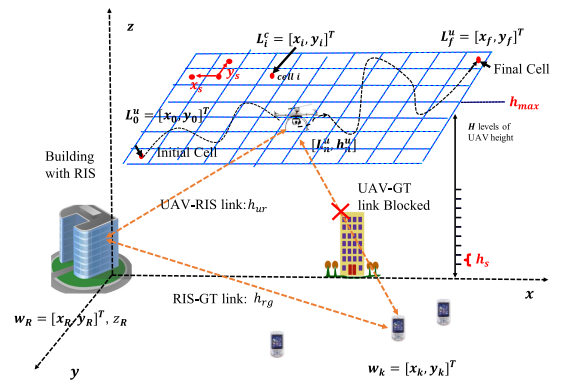


Fig. 1. RIS-assisted UAV communication system.

be approximated by $\{L_o^u, L_1^u, \dots, L_n^u, \dots, L_N^u, L_f^u\}$. For vertical dimension, assume that $h_n^u \in \mathcal{H} \triangleq \{1, 2, \dots, H\}$ is the level of the height of UAV in each time slot n and H is the total number of the levels of the height. Then the height of the UAV can be denoted as $H_n^u = h_n^u h_s$, where $h_s = \lfloor h_{\max}/H \rfloor$ is the distance of each height level of the UAV. Also assume that the height of UAV is constrained by $h_{\min} \leq H_n^u \leq h_{\max}$ with $h_{\min}$ and $h_{\max}$ being the minimum and maximum height respectively. In addition, we assume that t_n^u is the duration of time slot n and can be sufficiently small, i.e. $t_{\min} \leq t_n^u \leq t_{\max}$. Then the total mission completion time of the UAV can be given by $\tau = \sum_{n=1}^N t_n^u$. As a result, the UAV's trajectory can be represented by N waypoints in 3D coordinates as $[L_n^u, H_n^u], \forall n \in \mathcal{N}$, together with the duration of the time slots $t_n^u, \forall n \in \mathcal{N}$. Note that with the given maximum UAV horizontal speed $V_{\max}^h$, the number of time slots N can be chosen sufficiently large, such that the UAV's location change within each time slot t_n^u can be assumed to be negligible as compared to the link distances from the UAV to GTs.

Given above assumptions, the horizontal flying velocity of the UAV at time slot n can be denoted as $v_n^h = \frac{\|L_{n+1}^u - L_n^u\|}{t_n^u} \leq V_{\max}^h, \forall n \in \mathcal{N}$, where $V_{\max}^h$ is the maximum horizontal velocity. If $v_n^h = 0$, UAV then will hover at time slot n . On the other hand, the vertical flying velocity of the UAV at time slot n can be denoted as $v_n^v = \frac{\|H_{n+1}^u - H_n^u\|}{t_n^u} \leq V_{\max}^v, \forall n \in \mathcal{N}$, where $V_{\max}^v$ is the maximum vertical velocity of the UAV. With the horizontal and vertical velocities calculated, the propulsion energy of the rotary-wing UAV at each time slot can be denoted as

$$e_n^{\text{uav}} = t_n^u \left(P_0 \left(1 + \frac{3(v_n^h)^2}{U_{\text{tip}}^2} \right) + \frac{1}{2} d_0 \rho s G (v_n^h)^3 \right. \\ \left. + P_1 \left(\sqrt{1 + \frac{(v_n^h)^4}{4v_0^4}} - \frac{(v_n^h)^2}{2v_0^2} \right)^{\frac{1}{2}} + P_2 v_n^v \right) \quad (1)$$

where P_0 and P_1 are the constant blade profile power and induced power in hovering status respectively; P_2 is the constant descending/ascending power; U_{tip} is the tip speed of the rotor blade; v_0 is the mean rotor induced velocity in hover; d_0 and s are the fuselage drag ratio and rotor solidity respectively; ρ and G denote the air density and rotor disc area respectively [12].

As shown in Fig. 1, there is a RIS deployed on the surface of the territorial buildings to redirect the signal between UAV and GT to avoid N-LoS connections. Similar to [27], we assume that the RIS have $M_c \times M_r$ passive reflecting units (PRUs) to form a UPA. In specific, each column of the UPA has M_c PRUs with an equal distance of d_c meters and each row of the UPA consists of M_r PRUs with an equal distance of d_r meters. Each PRU can passively change its phase-shift with an independent reflection coefficient: $r_{m_r, m_c} = a e^{j\theta_{m_r, m_c}}, \forall m_r \in 1, 2, \dots, M_r$ and $\forall m_c \in 1, 2, \dots, M_c$. One has $a \in [0, 1]$ as the fixed reflection loss of the RIS and $\theta_{m_r, m_c} \in [-\pi, \pi]$ to be the phase shift inserted at PRU (m_r, m_c) . The location of the first element of the RIS is denoted as $w_R = [x_R, y_R]^T$ in horizontal dimension and z_R in vertical dimension. Assume there are $\mathcal{K} \triangleq \{1, 2, \dots, K\}$ GTs on the ground, each of which is in low mobility mode;

$w_k = [x_k, y_k]^T \in \mathbb{R}^{2 \times 1}$ is the location of each GT k , and D_k is the amount of data of GT k to be processed. The RIS ensures that the UAV-GT link could be replaced by the two UAV-RIS and RIS-GT links, and the UAV-RIS and RIS-GT links being all in LoS connections. However, it is possible that the signal reflecting of the RIS cannot guarantee the LoS connection between UAV and RIS, like when the UAV-RIS link being closely blocked by the ground obstacles. In this paper, we will not consider such pessimistic situation and leave it to the further work.

$$g_n^{ur} = \frac{\sqrt{\xi}}{d_n^{ur}} \left[1, e^{-j\frac{2\pi}{\lambda} d_r \phi_n^{ur} \psi_n^{ur}}, \dots, e^{-j\frac{2\pi}{\lambda} (M_r-1) d_r \phi_n^{ur} \psi_n^{ur}} \right]^T \\ \otimes \left[1, e^{-j\frac{2\pi}{\lambda} d_c \varphi_n^{ur} \psi_n^{ur}}, \dots, e^{-j\frac{2\pi}{\lambda} (M_c-1) d_c \varphi_n^{ur} \psi_n^{ur}} \right]^T \quad (2)$$

Under this deployment, the channel gain $g_n^{ur} \in \mathbb{C}^{M_r \times M_c}$ of the UAV-RIS link at the n -th time slot is denoted in (2).

In the equation, ξ is the path loss at the reference distance $D_0 = 1$ m; $d_n^{ur} = \sqrt{(H_n^u - z_R)^2 + (L_n^u - w_R)^2}$; $\phi_n^{ur} = \frac{x_k - x_R}{d_n^{ur}}$ and $\varphi_n^{ur} = \frac{y_R - y_i}{d_n^{ur}}$ represent the cosine and sine of the horizontal angle of arrival (AoA) of the signal at the RIS; $\psi_n^{ur} = \frac{H_n^u - z_R}{d_n^{ur}}$ represent the sine of the vertical AoA of the signal at the RIS; λ is the carrier wavelength. Here, we assume a far-field array response vector model at the RIS since $d_n^{ur} \gg \max\{M_r d_r, M_c d_c\}$. Similarly, the channel gain $g_k^{rg} \in \mathbb{C}^{M_r \times M_c}$ of the RIS to the k -th GT is given by

$$g_k^{rg} = \frac{\sqrt{\xi}}{d_k^{rg}} \left[1, e^{-j\frac{2\pi}{\lambda} d_r \phi_k^{rg} \psi_k^{rg}}, \dots, e^{-j\frac{2\pi}{\lambda} (M_r-1) d_r \phi_k^{rg} \psi_k^{rg}} \right]^T \\ \otimes \left[1, e^{-j\frac{2\pi}{\lambda} d_c \varphi_k^{rg} \psi_k^{rg}}, \dots, e^{-j\frac{2\pi}{\lambda} (M_c-1) d_c \varphi_k^{rg} \psi_k^{rg}} \right]^T \quad (3)$$

where $d_k^{rg} = \sqrt{(z_R)^2 + (w_R - w_k)^2}$; $\phi_k^{rg} = \frac{x_k - x_R}{d_k^{rg}}$ and $\varphi_k^{rg} = \frac{y_k - y_R}{d_k^{rg}}$ represent the cosine and sine of the horizontal angle of departure (AoD) of the signal to the k -th GT; $\psi_k^{rg} = \frac{z_R}{d_k^{rg}}$ represents the sine of the vertical AoD of the signal to the k -th GT; λ is the carrier wavelength. To this end, the channel gain achieved at GT k can be written as

$$g_{k,n}^{\text{urg}} = a(g_k^{rg})^T \cdot \Theta_n \cdot g_n^{ur} \quad (4)$$

where $\Theta_n = \text{diag}(\theta_n) \in \mathbb{C}^{M_r M_c \times M_r M_c}$ is the RIS reflection phase coefficient matrix, and $\theta_n = [e^{j\theta_{1,1}^n}, \dots, e^{j\theta_{M_r, M_c}^n}, \dots, e^{j\theta_{M_r, M_c}^n}]^T \in \mathbb{C}^{M_r M_c \times 1}$.

To evaluate the possibility that the direct UAV-GT link being blocked, we consider the air-to-ground channel model in urban environments of [16], where the blocking probability between the UAV and the GT k at time slot n is given by

$$p_{k,n} = 1 - \frac{1}{1 + a \exp \left(-b \left(\arctan \left(\frac{H_n^u}{d_k^{ug}} \right) - a \right) \right)} \quad (5)$$

where $d_k^{ug} = \sqrt{(H_n^u)^2 + (L_n^u - w_k)^2}$; a and b are constant values depending on the environment. Then, the achievable average channel gain and data rate at the k -th GT is given by

$$g_{k,n} = \underbrace{(1 - p_{k,n}) \frac{\xi}{(d_k^{ug})^2}}_{\text{UAV-GT}} + \underbrace{p_{k,n} g_{k,n}^{\text{urg}}}_{\text{UAV-RIS-GT}} \quad (6)$$

$$r_{k,n} = c_{k,n} B \log_2 \left(1 + \frac{P g_{k,n}}{B \sigma^2} \right) \quad (7)$$

where P is the fixed transmit power of the UAV, B is the bandwidth, and σ is the noise variance; $c_{k,n} = \{0, 1\}$ denotes GT k being scheduled or not. One has $\sum_{k=1}^K c_{k,n} \leq 1, \forall n$ denoting RIS at most serve one GT at a time slot i.e. in TDMA mode. The more complicated OFDMA mode supporting multiple GTs served by the RIS at one time slot, will be considered in future.

B. Problem Formulation

Let $\mathbf{L} = \{L_n^u, n \in \mathcal{N}\}$, $\mathbf{H} = \{h_n^u, n \in \mathcal{N}\}$, $\mathbf{C} = \{c_{k,n}, n \in \mathcal{N}\}$, $\mathbf{T} = \{t_n^u, n \in \mathcal{N}\}$ and $\Theta = \{\Theta_n, n \in \mathcal{N}\}$, we model the optimization problem as

$$\mathcal{P} : \min_{\mathbf{L}, \mathbf{H}, \mathbf{C}, \mathbf{T}, \Phi} \sum_{n=1}^N e_n^{\text{uav}} \quad (8)$$

$$\text{s.t. } c_{k,n} = \{0, 1\}, \sum_{k=1}^K c_{k,n} \leq 1, \forall n \in \mathcal{N} \quad (8.1)$$

$$\sum_{n=1}^N t_n^u r_{k,n} \geq D_k, \forall k \in \mathcal{K}; \quad (8.2)$$

$$v_n^h \leq V_{\max}^h, \forall n \in \mathcal{N}; \quad (8.3)$$

$$v_n^v \leq V_{\max}^v, \forall n \in \mathcal{N}; \quad (8.4)$$

$$h_{\min} \leq H_n^u \leq h_{\max}, \forall n \in \mathcal{N}; \quad (8.5)$$

$$t_{\min} \leq t_n^u \leq t_{\max}, \forall n; \quad (8.6)$$

where we aim to minimize the total propulsion energy consumption of the UAV among all the time slots; (8.1) indicates that only one GT links to the UAV at one time slot; (8.2) is the constraint to ensure that the data transferring of each task k : D_k being completed within the mission time of the UAV, which is to ensure that the UAV working in a acceptable energy efficiency; (8.3), (8.4) (8.5) are the constraints of the rotary-wing UAV. One can see that (8) is a non-convex problem, due to the complicity of $r_{k,n}$ referring the channel gains of UAV-RIS and RIS-GT links. Also the target function of $\mathcal{P}$ is un-trackable, because the UAV energy consumption model is complicated. To partially solve $\mathcal{P}$ using traditional method like SCA, it will cost enormous computation cost making it impossible to be online deployed. Due to this reason, next, we will propose the DRL-based solution to address problem $\mathcal{P}$.

III. PROPOSED SOLUTION

In this section, we propose the DDQN and DDPG based algorithms to solve $\mathcal{P}$, where an agent is applied to collect the state of RIS-assisted UAV system; find the best action with respect to the reward; and then deploy the selected action to the UAV and RIS. Specifically, we first take the location of the UAV at current time slot as the state. Then, the designed Deep Neural Networks (DNNs) learn and select the movement and flight time of the UAV and the schedule of GTs as the following action. Next, the selected action will be deployed and the phases

of the received signals from the UAV-RIS-GT links at the GTs will be aligned to obtain a closed-form phase-shift solution of the RIS. The phase-shift of the RIS then will maximize the received signal energy between the UAV and GT leading to the highest reward of the selected action. Finally, the system will be updated with respect to the selected action and the DNNs will be trained.

To better explain the DRL algorithms, in rest part of this section, we first model the state, action and reward of the DDQN and DDPG algorithms as well as the passive phase-shift of RIS, and then give details of these two algorithms.

A. DDQN Algorithm

1) *State, Action and Reward*: The key idea of DDQN is to model the UAV trajectory, flight time and GT scheduling choices in each time slot n as the discrete action $a(n)$, and use the Q-value of each action to decide whether $a(n)$ will be selected or not considering current state $s(n)$ of the RIS-assisted UAV. The Q-value of each action is defined as

$$Q(s(n), a(n)) = \mathbb{E} \left[\sum_{n'=n}^N \gamma r(s(n'), a(n')) \mid s(n), a(n) \right] \quad (9)$$

where $Q(s(n), a(n))$ estimates the expected discounted cumulative reward, and the discount factor $\gamma \in (0, 1]$. Given problem $\mathcal{P}$, the reward of the state-action pair $(s(n), a(n))$ at time slot n can be defined as

$$r(s(n), a(n)) = \sum_{k=1}^K \sum_{n'=1}^{n+1} \frac{t_n^u r_{k,n'}}{e_{n'}^{\text{uav}}} - p_0 \quad (10)$$

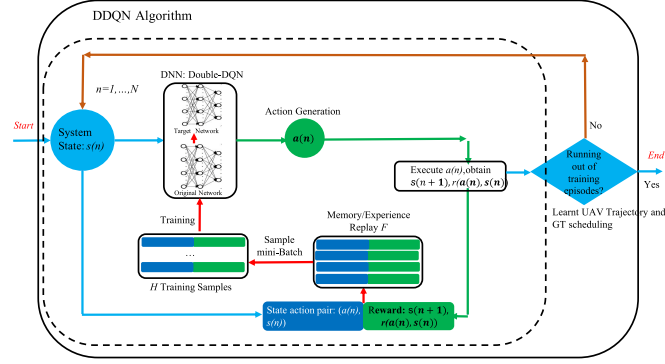
which is basically the ratio of the total amount of data of GTs up to time slot $n+1$, i.e. $\sum_{k=1}^K \sum_{n'=1}^{n+1} t_n^u r_{k,n'}$, to the UAV's propulsion energy, i.e., $\sum_{n'=1}^{n+1} e_{n'}^{\text{uav}}$. However, to guarantee (8.2), one will adopt a penalty p_0 to the reward, if the data transferred in time slot n being in a level lower than the average one, i.e. $t_n^u r_{k,n'} < \frac{D_k}{N}$.

With the Q-values defined in (9), the Q-learning process then adopts a policy to find the most proper action. In most cases, the greedy policy is adopted as

$$\pi(s(n)) = \arg \max_{a(n)} Q(s(n), a(n)) \quad (11)$$

where the optimal UAV trajectory, flight time and GT scheduling learnt by the DDQN algorithm are the action $a(n)$ having the highest Q-value with respect to state $s(n)$.

The state $s(n)$ of the RIS-assisted UAV at time slot n can be defined as $s(n) = \{s_u(n)\} \in \mathcal{S} \triangleq \mathcal{L} \times \mathcal{H}$, where $\mathcal{S}$ is the overall state space and $s_u(n)$ is the location of UAV, i.e., $s_u(n) = (L_n^u, H_n^u) \in \mathcal{L} \times \mathcal{H}$. Afterwards, we define $\mathcal{A}$ as the action space of the RIS-assisted UAV system, including locations, flight time of the UAV and the scheduling of GTs. Then the overall action at time slot n can be defined as $a(n) = (l_n, h_n, c_{k,n}, t_n^u) \in \mathcal{A} = \mathcal{L}_u \times \mathcal{H}_u \times \mathcal{C} \times \mathcal{T}$, where l_n and h_n being the UAV flying actions in horizontal and vertical dimensions that will be defined in (12) and (13). $\mathcal{T} = [t_{\min} : 0.1 \text{ ms} : t_{\max}]$ is the space of the discrete flight times, from where t_n^u will be chosen as the discrete value between $t_{\min}$ and $t_{\max}$ with 0.1 ms as the step size. $\mathcal{C} = \{c_{k,n}, \forall k, n\}$ is the action space of GT scheduling.

Fig. 2. DDQN Algorithm solving problem P .

Considering the flying actions of UAV, assume that the UAV only moves to one of the adjacent cells from its current cell during one time slot in horizontal dimension [32], and only to its adjacent height level in the vertical dimension. Then the UAV's horizontal location L_{n+1}^u in next time slot can be expressed as

$$L_{n+1}^u = L_n^u + l_n \quad (12)$$

where $l_n \in \mathcal{L}_u \triangleq \{(0, y_s), (0, -y_s), (x_s, 0), (-x_s, 0), (0, 0)\}$, with $\mathcal{L}_u$ being the horizontal action space of the UAV including moving to north, south, east, west, or remains at its location respectively. Considering vertical flying, the UAV's vertical location H_{n+1}^u in next time slot can be defined as:

$$H_{n+1}^u = H_n^u + h_n \quad (13)$$

where $h_n \in \mathcal{H}_u \triangleq \{h_s, -h_s, 0\}$, with $\mathcal{H}_u$ being the vertical action space of the UAV including ascending, descending or remaining at its current height respectively.

2) *DDQN Algorithm*: Based on the above defined state, action and reward, the DDQN procedure solving problem P is demonstrated as Fig. 2 below, which uses the techniques of target network [33], and experience replay [34].

In DDQN, we use DNN with weight vectors θ^Q to estimate the Q function: $Q(s(n), a(n)|\theta^Q)$ in (9). In such deep Q-learning process, the weight parameters θ^Q of the DNN is trained using the loss function as:

$$L(\theta^Q) = \mathbb{E}[y(n) - Q(s(n), a(n)|\theta^Q)] \quad (14)$$

where $y(n)$ is the target value, which can be estimated as:

$$y(n) = r(s(n), a(n)) + \gamma Q'(s(n+1), \arg \max_a Q(s(n+1)|\theta^Q)|\theta^{Q'}) \quad (15)$$

where the target value $y(n)$ is calculated following the double Q-learning approach and considering the Q-values of the original network $Q(\cdot)$ and the target networks $Q'(\cdot)$, i.e. $Q(s(n+1)|\theta^Q)$ and $Q'(s(n+1), \max_a Q(s(n+1)|\theta^Q)|\theta^{Q'})$ as defined respectively.

To this end, the DDQN procedure is defined as Algorithm 1 below, which composes two stages in each episode: exploration stage (steps 4 to 13) and training stage (steps 14 to 17). In the exploration stage, the 3D-trajectory and flight time of the UAV as well as GT's scheduling will be obtained either by a random

Algorithm 1: DDQN Algorithm.

```

1 Initialize the experience replay buffer  $F$ , time slot count  $N$ ;
2 Initialize  $\theta^Q$  and  $\theta^{Q'}$  of the original network  $Q(\cdot)$  and target
   network  $Q'(\cdot)$  respectively, and set  $\theta^{Q'} = \theta^Q$ ;
3 for episode  $= 1, \dots, E$  do
4   Set  $n = 1$ , initialize the system to state  $s(1)$ ;
5   while  $n = 1, \dots, N$  and task  $D_k$  is not completed do
6     Randomly select  $a(n) \in A(s(n))$  with probability  $\epsilon$ ,
       otherwise select action by  $a(n) = \pi(s(n))$ ;
7     if (UAV out of desired region) and (UAV exceeding
       horizontal/vertical velocity) then
8       Cancel the action and apply the penalty;
9     end
10    Apply the phase shift of the RIS;
11    Execute  $a(n)$  for  $s(n+1)$  and  $r(s(n), a(n))$ ;
12    Store sample  $(s(n), a(n), r(\cdot), s(n+1))$  in  $F$ ;
13  end
14  Select a random mini-batch of  $H$  samples
      $(s(j), a(j), r(j), s(j+1))$  from  $F$ ;
15  Calculate  $y(j) =$ 
      $r(j) + \gamma Q'(s(j+1), \max_a Q(s(j+1)|\theta^Q)|\theta^{Q'})$ ;
16  Update weights  $\theta^Q$  of  $Q(\cdot)$  by minimizing the loss:
      $L(\theta^Q) = \frac{1}{H} \sum_{j=1}^H (y(j) - Q(s(j), a(j)|\theta^Q))^2$ ;
17  Update the target networks:  $\theta^{Q'} = \delta \theta^Q + (1 - \delta) \theta^{Q'}$ ;
18 end

```

approach (with probability ϵ) or the greedy approach as defined in (11). If the selected l_n , h_n and t_n^u cause the UAV flying out of the defined region or the velocity exceeding the required level, i.e., $V_{\max}^h$ or $V_{\max}^v$, we set $l_n = (0, 0)$ or $h_n = 0$ to control the UAV to remain at its current location (step 8). Additionally, in the above cases, the penalty p will be applied to the correlated reward $r(s(n), a(n))$. Furthermore, with the obtained UAV's trajectory and the scheduling of GTs, the phase-shift of the RIS will be obtained at step 10, which will be discussed in detail later. Then, the next state $s(n+1)$ and corresponding reward $r(s(n), a(n))$ can be updated. Next, the newly generated sample $(s(n), a(n), r(\cdot), s(n+1))$ will be stored in the experience buffer F .

In the training stage, a random mini-batch of H samples will be selected from the experience buffer to train the original network $Q(\cdot)$ and the target networks $Q'(\cdot)$. Then the target value $y(j)$ will be used to update the weights θ^Q of the original network by minimizing the loss $L(\theta^Q)$ at step 16, which follows the general definition in (14). Next, the weights $\theta^{Q'}$ of the target network $Q'(\cdot)$ will be updated at step 17 but in a lower frequency than the original network $Q(\cdot)$ to avoid the overestimation issue. The DDQN based algorithm will terminate if running out of allowed episodes or each GT completing its task.

3) *Phase-Shift of the RIS*: As demonstrated in Algorithm 1, the passive phase-shift of the RIS is carried out within the DRL procedure to maximize the data transferring rate in each time slot. In theory, with the location of the UAV and the scheduled k -th GT available at time slot n , we can maximize the achievable rate of GT k by passively shift the corresponding phase of each

Algorithm 2: DDPG Algorithm.

```

1 Initialize the experience replay buffer  $F$ , time slot count  $N$ ;
2 Initialize  $\theta^\pi$  and  $\theta^{\pi'}$  of the original policy network  $\pi(\cdot)$  and
   target network  $\pi'(\cdot)$  respectively, and set  $\theta^{\pi'} = \theta^\pi$ ;
3 Initialize  $\theta^Q$  and  $\theta^{Q'}$  of the original network  $Q(\cdot)$  and target
   network  $Q'(\cdot)$  respectively, and set  $\theta^{Q'} = \theta^Q$ ;
4 for episode = 1,...,  $E$  do
5   Set  $n = 1$ , initialize the system to state  $s(1)$ ;
6   while  $n = 1, \dots, N$  and task  $D_k$  is not completed do
7     Select select action by  $a(n) = \pi(s(n)|\theta^\pi) + \mathcal{R}$ ,
       where  $\mathcal{R}$  the random funtion;
8     if (UAV out of desired region) and (UAV exceeding
       horizontal/vertical velocity) then
9       | Cancel the action and apply the penalty;
10    end
11    Apply the phase shift of the RIS;
12    Execute  $a(n)$  for  $s(n+1)$  and  $r(s(n), a(n))$ ;
13    Store sample  $(s(n), a(n), r(\cdot), s(n+1))$  in  $F$ ;
14  end
15  Select a random mini-batch of  $H$  samples
     $(s(j), a(j), r(j), s(j+1))$  from  $F$ ;
16  Calculate
     $y(j) = r(j) + \gamma Q'(s(j+1), \pi'(s(j+1)|\theta^{\pi'})|\theta^{Q'})$ ;
17  Update weights  $\theta^Q$  of  $Q(\cdot)$  by minimizing the loss:
     $L(\theta^Q) = \frac{1}{H} \sum_{j=1}^H (y(j) - Q(s(j), a(j)|\theta^Q))^2$ ;
18  Update the target networks:  $\theta^{Q'} = \delta \theta^Q + (1 - \delta) \theta^{Q'}$ ;
19  Update weights  $\theta^\pi$  of  $\pi(\cdot)$  by applying gradient:
     $\nabla_{\theta^\pi} J(\pi) \approx \frac{1}{H} \sum_i^H (\nabla_a Q(s, a|\theta^Q)|_{s=s(i), a=\pi(s(i))} \nabla_{\theta^\pi} \pi(s|\theta^\pi)|_{s(i)})$ ;
20  Update the target networks:  $\theta^{\pi'} = \delta \theta^\pi + (1 - \delta) \theta^{\pi'}$ ;
21 end

```

IV. PERFORMANCE EVALUATION AND ANALYSIS

In this section, we validate the proposed DDQN and DDPG algorithms to the RIS-assisted UAV system (UAV-RP) through simulation and compare the numerical results with benchmark systems. One of the benchmark system works on the UAV without RIS (UAV/R), and the other one works on the UAV with RIS but without the optimal passive phase-shift (UAV-R/P). Both of the two benchmark systems optimize the UAV trajectory and GT scheduling by using the DDQN method. The simulation setting on RIS-assisted UAV system is listed in Table II.

We consider the GTs are randomly distributed in the targeted area and set $L_o^u = [0, 0]^T$, $w_R = [50, 50]^T$, $H = 200$ m, $z^R = 50$, $d = \frac{\lambda}{2}$, $F = 3200$ and $H=32$. We set the air-ground communications following the scenario discussed in [16], and the propulsion model of the rotary-wing UAV based on the settings in [12]. The simulation is conducted in Python 2.7 and Tensorflow 1.14.0 [35] to implement the DNNs both in the DDQN and DDPG algorithms. In DDQN, the original and target networks employ 2 layers and each layer has 20 neurons and we utilize ReLu as the activation function and RMSProp as the optimizer to train the DNNs. The parameters of each DNNs are randomly initialized following a zero-mean normal distribution. Based on above settings, one can find the action space of the DDQN algorithm is $|\mathcal{L}_u| \times |\mathcal{H}_u| \times K \times \frac{t_{max} - t_{min}}{0.1 \text{ ms}} =$

TABLE II
PARAMETER SETTINGS ON SIMULATION

Parameter	Value
Bandwidth B , UAV power P	2MHz, 5mW
Blocking parameters a, b	9.61, 0.16 [16]
$M; \theta_i[1]$	20, 0°
$U_{tip}, v_0, d_0, s, \rho, G$	120, 4.3, 0.6, 0.05, 1.225, 0.503 [12]
P_0, P_1, P_2	$\frac{12 \times 30^3 \times 0.4^3}{8} \rho s G, \frac{1.1 \times 20^{3/2}}{\sqrt{2} \rho G}, 11.46$
V_{max}^h, V_{max}^v	10m/s, 10m/s
GTs: K , Task: D_k (scenario 1)	6, 256~512 Kb
GTs: K , Task: D_k (scenario 2)	6, 512~1024 Kb
t_{min}, t_{max} , Noisy density: σ	1s, 3s, -169dBm/Hz
Flying height: h_0^u, h_{min}, h_{max}	100m, 30m, 100m
Time slots and episodes: N, E	600, 60
Area size, Number of Cells C :	1000m*1000m, 10000

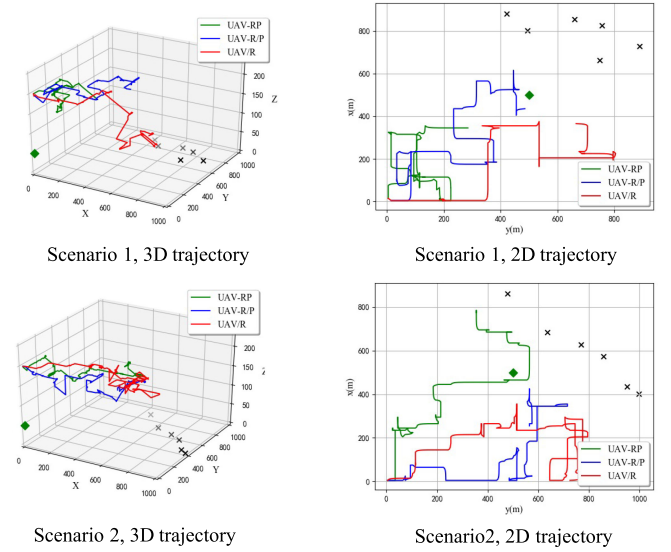


Fig. 4. UAV trajectories of different systems in 2D and 3D.

$5 \times 3 \times 6 \times 21 = 1890$. In DDPG, the original and target networks of the policy network both employ 2 layers DNNs. The first layer has 30 neurones and utilizes ReLu as the activation function, and the second layer utilizes Tanh as the activation function. We apply AdamOptimizer to train the DNNs of the policy network. Meanwhile, the original and target networks of the Q-network also employ 2 layers DNNs. Each layer has 30 neurones and utilizes ReLu as the activation function. We apply AdamOptimizer to train the DNNs of the Q-network. The parameters of each DNNs are randomly initialized following a zero-mean normal distribution.*

In Fig. 4, we plot the 3D trajectories and their projections in 2D planes of different systems. One observes that the UAV-RP system normally adopts the shortest distance to save the propulsion energy, as compared to the cases of UAV/R and UAV-R/P. In UAV/R case, the UAV flies very close to each GT and dives to a lower altitude so as to establish communication links to GTs,

*The simulation code can be obtained: <https://github.com/HaiboMei/UAV-RIS-DRL-git>

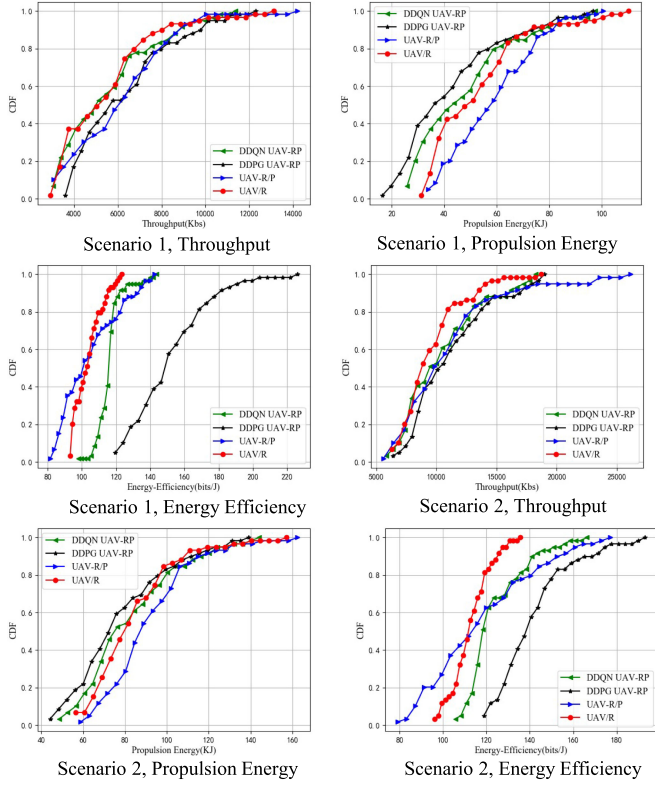


Fig. 5. The performance on propulsion energy, throughput and energy efficiency led by different solutions.

TABLE III
NUMERICAL RESULTS

Scenarios 1	Propulsion Energy (KJ)	Energy-efficiency (bits/J)
DDQN UAV-RP	48.85	116.85
DDPG UAV-RP	43.71	153.72
UAV-R/P	59.66	105.47
UAV/R	53.78	104.07
Scenarios 2	Propulsion Energy (KJ)	Energy-efficiency (bits/J)
DDQN UAV-RP	84.06	125.08
DDPG UAV-RP	79.44	143.24
UAV-R/P	93.77	118.88
UAV/R	86.28	113.12

which obviously result in low energy-efficiency performance. On the contrary, the UAV in UAV-RP system normally tries to approach to the RIS to find the suitable trajectories to serve the users.

In Fig. 5, we further depict the cumulative distribution function (CDF) of the propulsion energy, throughput, energy-efficiency performance in three compared systems. One can see that the UAV-RP with DDQN and DDPG algorithm always have the relatively high performance in both tested scenarios, followed by UAV-R/P and UAV/R. Additionally, the DDPG algorithm gives the best performance overall. The numerical data related to Fig. 5 is listed as Table III, where the average propulsion energy and energy-efficiency of the UAV are demonstrated with respect to all the considered situations.

V. CONCLUSION

This paper has proposed the DDQN and DDPG based solutions to jointly optimize the 3D trajectory of UAV and the phase-shift of RIS. The proposed algorithm has been proved effective to improve the energy-efficiency of the UAV, while satisfying data transmission requirement of GTs. Future work will focus on more complex scenario where multiple UAVs are deployed and multiple antennas are installed at UAVs to further improve the performance of the system.

REFERENCES

- [1] Y. Zeng, R. Zhang, and T. J. Lim, "Wireless communications with unmanned aerial vehicles: Opportunities and challenges," *IEEE Commun. Mag.*, vol. 54, no. 5, pp. 36–42, May 2016.
- [2] C. Liu, W. Feng, Y. Chen, C. X. Wang, and N. Ge, "Cell-free satellite-uav networks for 6G wide-area Internet of Things," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 4, pp. 1116–1131, Apr. 2021.
- [3] D. G.C., A. Ladas, Y. A. Sambo, H. Pervaiz, C. Politis, and M. A. Imran, "An overview of post-disaster emergency communication systems in the future networks," *IEEE Wireless Commun.*, vol. 26, no. 6, pp. 132–139, Dec. 2019.
- [4] Y. Zeng, Q. Wu, and R. Zhang, "Accessing from the sky: A tutorial on uav communications for 5G and beyond," *Proc. IEEE*, vol. 107, no. 12, pp. 2327–2375, Dec. 2019.
- [5] L. Liu, S. Zhang, and R. Zhang, "Comp in the sky: UAV placement and movement optimization for multi-user communications," *IEEE Trans. Commun.*, vol. 67, no. 8, pp. 5645–5658, Aug. 2019.
- [6] J. Zhang, Y. Zeng, and R. Zhang, "UAV-enabled radio access network: Multi-mode communication and trajectory design," *IEEE Trans. Signal Process.*, vol. 66, no. 20, pp. 5269–5284, Oct. 2018.
- [7] S. Zhang, Y. Zeng, and R. Zhang, "Cellular-enabled UAV communication: A connectivity-constrained trajectory optimization perspective," *IEEE Trans. Commun.*, vol. 67, no. 3, pp. 2580–2604, Mar. 2019.
- [8] Y. Zeng, J. Lyu, and R. Zhang, "Cellular-connected UAV: Potential, challenges, and promising technologies," *IEEE Wireless Commun.*, vol. 26, no. 1, pp. 120–127, Feb. 2019.
- [9] A. Al-Hourani and K. Gomez, "Modeling cellular-to-UAV path-loss for suburban environments," *IEEE Wireless Commun. Lett.*, vol. 7, no. 1, pp. 82–85, Feb. 2018.
- [10] Y. Zeng and R. Zhang, "Energy-efficient UAV communication with trajectory optimization," *IEEE Trans. Wireless Commun.*, vol. 16, no. 6, pp. 3747–3760, Jun. 2017.
- [11] Q. Wu and R. Zhang, "Common throughput maximization in UAV-enabled OFDMA systems with delay consideration," *IEEE Trans. Commun.*, vol. 66, no. 12, pp. 6614–6627, Dec. 2018.
- [12] H. Mei, K. Wang, D. Zhou, and K. Yang, "Joint trajectory-task-cache optimization in UAV-enabled mobile edge networks for cyber-physical system," *IEEE Access*, vol. 7, pp. 156 476–156 488, 2019.
- [13] S. Boyd and L. Vandenberghe, *Convex Optimization*. New York, Cambridge Univ. Press, 2004, pp. 267–288.
- [14] V. Mnih *et al.*, "Human-level control through deep reinforcement learning," *Nature*, vol. 518, no. 7540, pp. 529–533, 2015. [Online]. Available: <https://doi.org/10.1038/nature14236>
- [15] Q. Wu, L. Liu, and R. Zhang, "Fundamental trade-offs in communication and trajectory design for UAV-enabled wireless network," *IEEE Wireless Commun.*, vol. 26, no. 1, pp. 36–44, Feb. 2019. [Online]. Available: <http://dx.doi.org/10.1109/MWC.2018.1800221>
- [16] A. Al-Hourani, S. Kandeepan, and S. Lardner, "Optimal LAP altitude for maximum coverage," *IEEE Wireless Commun. Lett.*, vol. 3, no. 6, pp. 569–572, Dec. 2014.
- [17] Y. Zeng, J. Xu, and R. Zhang, "Energy minimization for wireless communication with rotary-wing UAV," *IEEE Trans. Wireless Commun.*, vol. 18, no. 4, pp. 2329–2345, Apr. 2019.
- [18] M. Di Renzo *et al.*, "Smart radio environments empowered by reconfigurable intelligent surfaces: How it works, state of research, and the road ahead," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 11, pp. 2450–2525, Nov. 2020.
- [19] C. Huang, A. Zappone, G. C. Alexandropoulos, M. Debbah, and C. Yuen, "Reconfigurable intelligent surfaces for energy efficiency in wireless communication," *IEEE Trans. Wireless Commun.*, vol. 18, no. 8, pp. 4157–4170, Aug. 2019.

- [20] S. Hu, F. Rusek, and O. Edfors, "Beyond massive MIMO: The potential of data transmission with large intelligent surfaces," *IEEE Trans. Signal Process.*, vol. 66, no. 10, pp. 2746–2758, May 2018.
- [21] Q. Wu and R. Zhang, "Intelligent reflecting surface enhanced wireless network via joint active and passive beamforming," *IEEE Trans. Wireless Commun.*, vol. 18, no. 11, pp. 5394–5409, Nov. 2019.
- [22] Y. Han, W. Tang, S. Jin, C. K. Wen, and X. Ma, "Large intelligent surface-assisted wireless communication exploiting statistical CSI," *IEEE Trans. Veh. Technol.*, vol. 68, no. 8, pp. 8238–8242, Aug. 2019.
- [23] T. Bai, C. Pan, Y. Deng, M. ElKashlan, A. Nallanathan, and L. Hanzo, "Latency minimization for intelligent reflecting surface aided mobile edge computing," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 11, pp. 2666–2682, Nov. 2020.
- [24] C. Huang, R. Mo, and C. Yuen, "Reconfigurable intelligent surface assisted multiuser mimo systems exploiting deep reinforcement learning," *IEEE J. Sel. Areas Commun.*, vol. 38, no. 8, pp. 1839–1850, Aug. 2020.
- [25] H. Long, M. Chen, Z. Yang, B. Wang, X. Y. Z. Li, and M. Shikh-Bahaei, "Reflections in the sky: Joint trajectory and passive beamforming design for secure UAV networks with reconfigurable intelligent surface," *IEEE Trans. Wireless Commun.*, vol. 20, no. 10, pp. 6402–6417, Oct. 2021, doi: [10.1109/TWC.2021.3073746](https://doi.org/10.1109/TWC.2021.3073746).
- [26] M. Hua, L. Yang, Q. Wu, C. Pan, C. Li, and A. Lee Swindlehurst, "UAV-assisted intelligent reflecting surface symbiotic radio system," *IEEE Trans. Wireless Commun.*, vol. 20, no. 9, pp. 5769–5785, Sep. 2021.
- [27] Z. Wei *et al.*, "Sum-rate maximization for IRS-assisted UAV OFDMA communication systems," *IEEE Trans. Wireless Commun.*, vol. 20, no. 4, pp. 2530–2550, Apr. 2021.
- [28] L. Sixian, D. Bin, M. D. Renzo, T. Meixia, and Y. Xiaojun, "Robust secure UAV communications with the aid of reconfigurable intelligent surfaces," 2020. [Online]. Available: <https://arxiv.org/abs/2008.09404>
- [29] S. Li, B. Duo, X. Yuan, Y. Liang, and M. Di Renzo, "Reconfigurable intelligent surface assisted UAV communication: Joint trajectory design and passive beamforming," *IEEE Wireless Commun. Lett.*, vol. 9, no. 5, pp. 716–720, May 2020.
- [30] X. Liu, Y. Liu, and Y. Chen, "Machine learning empowered trajectory and passive beamforming design in UAV-RIS wireless networks," *IEEE J. Sel. Areas Commun.*, vol. 39, no. 7, pp. 2042–2055, Jul. 2021.
- [31] L. Wang, K. Wang, C. Pan, W. Xu, and N. Aslam, "Joint trajectory and passive beamforming design for intelligent reflecting surface-aided UAV communications: A deep reinforcement learning approach," 2020. [Online]. Available: <https://arxiv.org/abs/2007.08380>
- [32] M. A. Abd-Elmagid, A. Ferdowsi, H. S. Dhillon, and W. Saad, "Deep reinforcement learning for minimizing age-of-information in UAV-assisted networks," in *Proc. IEEE Glob. Commun. Conf.*, 2019, pp. 1–6.
- [33] L. Huang, S. Bi, and Y. J. Zhang, "Deep reinforcement learning for on-line computation offloading in wireless powered mobile-edge computing networks," *IEEE Trans. Mobile Comput.*, vol. 19, no. 11, pp. 2581–2593, Nov. 2020.
- [34] C. H. Liu, Z. Chen, J. Tang, J. Xu, and C. Piao, "Energy-efficient UAV control for effective and fair communication coverage: A deep reinforcement learning approach," *IEEE J. Sel. Areas Commun.*, vol. 36, no. 9, pp. 2059–2070, Sep. 2018.
- [35] M. Abadi *et al.*, "Tensorflow: A system for large-scale machine learning," *USENIX Association*, 2016.

Haibo Mei received the B.Sc. and M.Sc. degrees from the School of Computer Science and Engineering, University of Electronic Science and Technology of China, Chengdu, China, in 2005 and 2008, respectively, and the Ph.D. degree from the School of Electronic Engineering and Computer Science, Queen Mary University of London (QMUL), London, U.K., in 2012. He was a Postdoctoral Research Assistant with QMUL and a Senior R&D Engineer with Securus Software Ltd, U.K. He is currently a Lecturer with the University of Electronic Science and Technology of China. His research interests include resource efficiency and self-organization of wireless communications, UAV communication, and mobile cloud computing.

Kun Yang (Senior Member, IEEE) received the B.Sc. and M.Sc. degrees from Computer Science Department, Jilin University, Changchun, China, and the Ph.D. degree from the Department of Electronic and Electrical Engineering, University College London, London, U.K. He is currently a Chair Professor with the School of Computer Science and Electronic Engineering, University of Essex, Colchester, U.K., leading the Network Convergence Laboratory, U.K. He is also an Affiliated Professor with the University of Electronic Science and Technology of China, Chengdu, China. He has authored or coauthored more than 200 papers. His research interests include wireless networks, network convergence, future internet technology, data and energy cooperation, and mobile computing. He manages research projects funded by various sources such as U.K. EPSRC, EU FP7/H2020, and industries. He is on the editorial boards of both IEEE and non-IEEE journals.

Qiang Liu (Member, IEEE) received the B.Sc., M.Sc., and Ph.D. degrees from the School of Information and Communication, University of Electronic Science and Technology of China, Chengdu, China, in 1996, 2000, and 2012, respectively. He is currently an Associate Professor with the University of Electronic Science and Technology of China. His main research interests include wireless communication networks, Internet of Things, low power WAN, and molecular communication. He has been a Member of IEEE ComSoc.

Kezhi Wang (Senior Member, IEEE) received the B.E. and M.E. degrees from the College of Automation, Chongqing University, Chongqing, China, in 2008 and 2011, respectively, and the Ph.D. degree from the University of Warwick, Coventry, U.K., in 2015. He was a Senior Research Officer with the University of Essex, Colchester, U.K. He is currently a Lecturer with the Department of Computer and Information Sciences, Northumbria University, Newcastle upon Tyne, U.K. His research interests include wireless communication, signal processing, and mobile cloud computing.